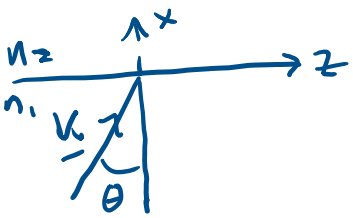
1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value. [20]

a) An optical plane wave is incident on a planar interface at an angle of 30° from the normal. The free space wavelength is 1.30 μm, and the refractive index in the incident medium $n_1 = 1.45$. Calculate the magnitude of the component of the wave vector k along the interface.



Direction along the interface = z

$k_z = |k| \sin\theta \quad \theta = 30° \quad \sin\theta = \frac{1}{2}$

$k_z = \frac{2\pi}{1.3\times10^{-6}} \times \frac{1}{2} = 2.42\times10^{6}\ m^{-1}$

b) An optical plane wave is incident on a planar interface at exactly the critical angle. Show that the phase change on reflection will be 0°.

The reflection coefficient $\frac{E_r}{E_i} = \frac{k_{1x} - k_{2x}}{k_{1x} + k_{2x}}$

At the critical angle, $k_{2x} = 0$

Then $\frac{E_r}{E_i} = 1$ ∴ no phase change

c) A certain single mode fibre has effective index n′ = 1.48 at the operating wavelength, and carries a signal of data rate 100 Mbit/s. Calculate the number of bits per meter of fibre for the propagating signal.

Time to propagate 1 m $\tau = \frac{1\,m}{v_p} = \frac{1 \times n'}{c} = \frac{1.48\ m}{3\times10^8\ m/s} = 4.93\ ns$

$\frac{1}{B} = 10\ ns$

$\dot{N} = 0.493\ bits/m$

d) An optical signal of power level 5.0 mW is launched into a fibre of length 25 km. The power level arriving at the other end of the fibre is 0.2 mW. Find the fibre attenuation coefficient in dB/km.

Attenuation in dB $= 10 \log\left(\frac{5}{0.2}\right) = 14.0\ dB$

$\alpha_{dB} = \frac{14}{25} = 0.56\ dB/km$

e) An optical fibre is found to have an attenuation coefficient of 1.5 dB/km at a wavelength λ_1 sufficiently below the minimum attenuation wavelength that Rayleigh scattering is the dominant loss mechanism. Estimate the attenuation coefficient at a wavelength $\lambda_2 = \lambda_1/2$.

Rayleigh attenuation is prop to $\frac{1}{\lambda^4}$, so halving λ means 16× the attenuation rate.

Then $\alpha_{dB} = 16 \times 1.5 = 24\ dB/km$

f) An optical signal at 1.5 Gbit/s generates a photocurrent in the photodetector of 350 nA. Assuming that the quantum efficiency of the detector $\eta = 1$, calculate the number of photons received per bit.

If $\eta = 1$, no. of photons/sec = no of electrons/sec in the photocurrent $\dot{N} = \frac{350 \times 10^{-9}}{1.6 \times 10^{-19}} = 219 \times 10^{10}$

No. of photons/bit $= \frac{\dot{N}}{B} = \frac{219 \times 10^{10}}{1.5 \times 10^{9}} = 1460$

g) A laser diode operating at a nominal wavelength of 1.52 μm has a quantum efficiency $\eta = 0.85$. What is its slope efficiency?

$S = \eta \frac{hc}{e\lambda} = \frac{0.85 \times 6.63 \times 10^{-34} \times 3 \times 10^{8}}{1.6 \times 10^{-19} \times 1.52 \times 10^{-6}} = 0.695$ W/A

h) A certain unamplified optical link using a standard p-i-n photodiode detector has equal levels of shot noise and thermal noise. If the fibre length is doubled, which noise source would you expect to now dominate?

Increased length will reduce the received power, and will reduce the shot noise, but will not affect the receiver noise, so receiver noise will now dominate.

i) Estimate a suitable thickness and index be for an anti-reflection coating on the end of a silica fibre emitting into air, if the free space wavelength is 1330 nm?

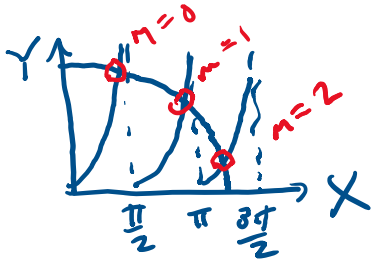
Estimate the mode index as $n' = 1.46$, then $n_{AR} \simeq \sqrt{1.46 \times 1}$ $= 1.21$

Thickness $t = \frac{\lambda}{4} = \frac{\lambda_0}{4 n_{AR}} \simeq \frac{1330}{4 \times 1.21} = 275$ nm

j) A certain optical fibre supports two modes of effective indices 1.455 and 1.465 respectively. Which of the two modes will experience higher loss due to bending of the fibre? Give a brief explanation.

The mode with the lower n' (1.455) has more light in the cladding, so is less strongly guided and will experience higher bending loss

2.a) A symmetric slab waveguide as shown in Fig. 2.1 has a core thickness d = 10.0 μm, a core refractive index n_1 = 1.520, and cladding refractive index n_2 = 1.510. It is operated at a nominal wavelength of λ_o = 1.45 μm. Find the number of TE modes supported by the waveguide. Illustrate your solution by sketching an X-Y plot, where $X = k_{1x}d/2$ and $Y = \kappa d/2$. [8]



$NA = \sqrt{1.52^2 - 1.51^2} = .174$

$R = \frac{NA \cdot d \cdot \pi}{\lambda_o}$ $R = \frac{.174 \times 10 \times \pi}{1.45} = 1.2\pi$

Three modes are supported:
$m = 0, 1, 2.$

b) For the waveguide described in (a), calculate to 3 decimal places the effective index n′ of each of the supported TE modes. [6]

For $m=0$ use $Y = X \tan X$, with $X^2 + Y^2 = R^2$

gives $\frac{\cos X}{X} = \frac{1}{R} = 0.2652$

Successive approximation gives $X = 1.236$

$n_0' = \sqrt{n_1^2 - \left(\frac{2X}{kd}\right)^2} = 1.519$

For $m=1$ use $Y = -X \cot X$, $\frac{\sin X}{X} = \frac{1}{R} = 0.2652$

successive approx: $X = 2.44$

$n_1' = 1.516$

For $m=2$ $\cos X$ will be negative so $\frac{\cos X}{X} = -.2652$

Get initial estimate from plot, $X \cong 3.3$

Successive approx $X = 3.51$, $n_2' = 1.511$

c) Light is launched in the waveguide described in (a), into all modes. If at the entrance to the waveguide z = 0 the m=0 and m=1 modes are in phase, what will be the propagation distance Δz before they are 180° out of phase? [6]

Phase change $\Delta\varphi = k\Delta x = n' k_0 \Delta z$

$\Delta\varphi_0 - \Delta\varphi_1 = k_0 (n_0' - n_1') \Delta z = \pi$

$\Delta z = \frac{\pi}{2\pi} \lambda_0 \cdot \frac{1}{1.519 - 1.516} = \frac{0.725\,\mu m}{0.003} = 242\,\mu m$

3. a) An optical link has a transmitted power Φ_T, a fibre loss coefficient α, a receiver with noise equivalent power NEP, and a required signal-to-noise ratio SNR. The fibre has a dispersion coefficient D, and the transmitting laser has spectral width σ_λ.
Derive expressions for the maximum bit rate B as a function of fibre length L, for each of two cases:
A: Considering receiver noise as the only noise source; and
B: By neglecting noise and assuming B is limited by a requirement to keep the dispersion penalty below 3 dB. [6]

A: $SNR = \frac{\Phi_T e^{-\alpha L}}{NEP \cdot \sqrt{\Delta f}}$ $\quad \Delta f = B/2$

$\sqrt{\frac{B}{2}} = \frac{\Phi_T e^{-\alpha L}}{NEP \cdot SNR}$ $\quad B = 2\left(\frac{\Phi_T}{NEP \cdot SNR}\right)^2 e^{-2\alpha L}$

B: need to consult notes: For a dispersion penalty of 3 dB, $BLD\sigma_\lambda \approx 0.22$ (Fig 10.3)

Then $B = \frac{0.22}{LD\sigma_\lambda}$

b) For the link described in (a) and the parameter values below, sketch a single graph showing log(B) vs L for each of the two cases A and B. Using this graph, estimate the range of lengths L for which the dispersion limit of Case B is more demanding. [6]
$\Phi_T = 6$ mW, $\alpha_{dB} = 0.3$ dB/km, SNR = 12, NEP = 10 pW/$\sqrt{\text{Hz}}$
$\sigma_\lambda = 2.0$ nm, D = 5 ps/nm·km

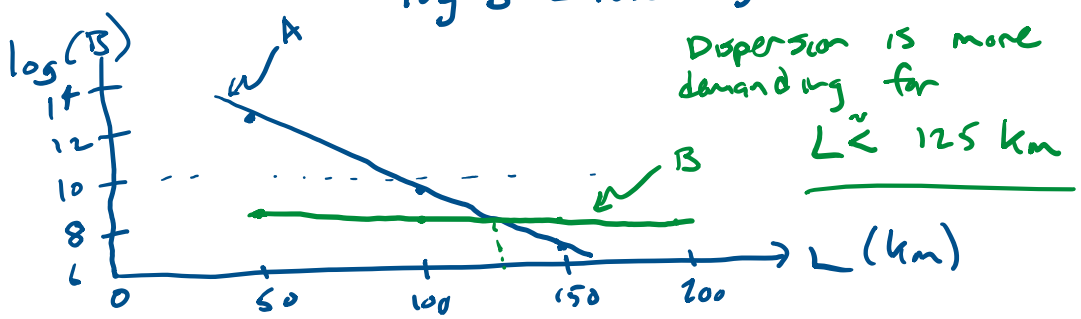
For Case A, $\left(\frac{\Phi_T}{NEP \cdot SNR}\right)^2 = \left(\frac{6\times10^{-3}}{10^{-11}\cdot 12}\right)^2 = \frac{10^{16}}{4}$

$\log B = \log(0.5\times10^{16}) - 2\alpha L(\log e)$ $\quad \alpha = \frac{\alpha_{dB}}{10\log e}$

$= 15.7 - \frac{\alpha_{dB}}{5} L = 15.7 - 0.06L$

For Case B: $B = \frac{0.22}{5\times10^{-12}\times 2}\cdot\frac{1}{L} = \frac{2.2\times10^{10}}{L}$

$\log B = 10.3 - \log L$



Dispersion is more demanding for $L \lesssim 125$ km

c) Assuming a fibre link is limited only by receiver noise, i.e. that other noise sources and dispersion can be neglected, derive an expression for the length L for which the bit rate-length product BL is maximised. Explain why the bit rate in such a case is unlikely to be achievable, using a quantitative example. [4]

$B = 2\left(\frac{\Phi_T}{NEP \cdot SNR}\right)^2 e^{-2\alpha L}$ in this case

$= J$ for convenience

$B \cdot L = J \cdot L e^{-2\alpha L}$

$\frac{\partial (BL)}{\partial L} = J e^{-2\alpha L} - 2\alpha J L e^{-2\alpha L} = 0$

$2\alpha L = 1 \qquad L = \frac{1}{2\alpha}$

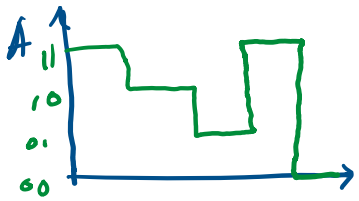
In this case, $B = \frac{2}{e}\left(\frac{\Phi_T}{NEP \cdot SNR}\right)^2$

For the parameters in (b):

$B = \frac{2}{2.72}\left(\frac{6\times10^{-3}}{10^{-11}\times 12}\right)^2 = \frac{10^{16}}{5.4} \approx 2000 \frac{Tbit}{s}$

This is many orders of magnitude higher than the bandwidths of sources, detectors or related electronics.

d) In a certain optical link, on-off keying is substituted with an amplitude shift keying modulation scheme having 4 equally spaced levels. Assuming the performance is limited by receiver noise in both cases, and that the total data rate is unchanged, estimate the quantitative effect on the signal-to-noise ratio. [4]

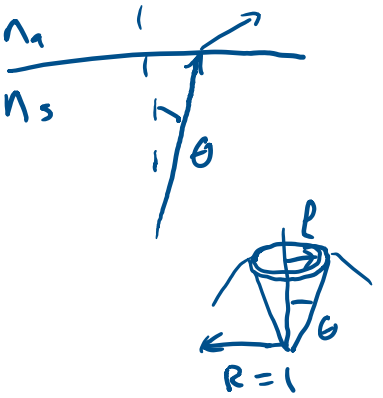


With 4 levels, the spacings are $\frac{1}{3}$ of the value with only 2 levels, so SNR is effectively reduced by 3×.

However, this scheme gives 2 bits/symbol, so Δf is halved. Since $SNR \propto \frac{1}{\sqrt{\Delta f}}$, this gives us a benefit of $\times\sqrt{2}$

So finally $SNR_4 \approx SNR_2 \times \frac{\sqrt{2}}{3}$ (roughly halved)

4. a) A light emitting diode (LED) has a horizontal active region located 6 μm below the flat, horizontal semiconductor-air interface. Calculate the maximum angle from the vertical, θ_m, for which emitted photons are able to cross the semiconductor-air interface, and find the fraction of photons which are emitted within this angular range. Assume that the active region emits photons equally in all directions. The refractive index of the semiconductor is 3.6. [6]



photons only cross the boundary if $\theta < \theta_c = \sin^{-1}\left(\frac{n_a}{n_s}\right)$

$\sin\theta_c = \left(\frac{1}{3.6}\right) \longrightarrow \theta_{max} = \sin^{-1}\left(\frac{1}{3.6}\right) \cong 16°$

to get fraction within θ_c, integrate annular segments.

$dA = 2\pi\rho\, d\theta = 2\pi \sin\theta\, d\theta \quad (R=1)$

$$f = \int_{\theta=0}^{\theta_c} dA \Big/ \int_0^{\pi/2} dA = \int_0^{\theta_c} \sin\theta\, d\theta \Big/ \int_0^{\pi/2} \sin\theta\, d\theta$$

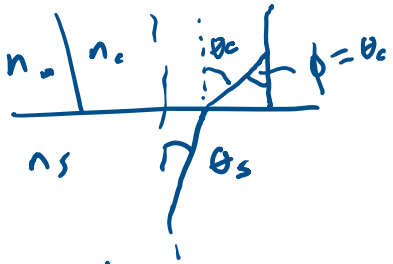
$$= \frac{1-\cos\theta_c}{1} \cong 1 - (1 - \theta_c^2/2) = \frac{\theta_c^2}{2} \quad (\theta_c \ll 1)$$

here θ_c = critical angle

$f \cong \frac{1}{2}\left(\frac{1}{3.6}\right)^2 = 0.039$

Taking into account also the downward photons, $f = .0195 \cong 2\%$

b) The LED described in (a) is now used to launch light into a multi-mode fibre, with a core diameter much larger than the width of the emitting region. The fibre is placed directly against the semiconductor surface, correctly centred and aligned, and has numerical aperture NA = 0.3. Calculate the maximum angular range from normal θ_{m2} for which emitted photons are able to enter and be guided within the fibre, and find the fraction of photons which are emitted within this angular range. State any other assumptions or approximations made. [4]



here θ_c = angle in core of fibre

We need TIR at the core-cladding boundary, $\cos\phi < n_a/n_c$

$\sin\theta_c < \sqrt{1-(n_a/n_c)^2}$

$< \frac{\sqrt{n_c^2 - n_a^2}}{n_c} = \frac{NA}{n_c}$

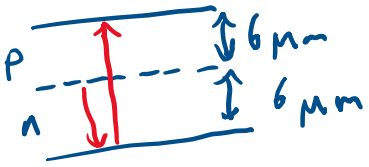
$\sin\theta_s = \frac{n_c}{n_s}\sin\theta_c$

$\sin\theta_s < \frac{NA}{n_s} = \frac{0.3}{3.6} = .083 \quad \theta = 4.76°$

$f \cong \frac{1}{2}\left(\frac{NA}{n_s}\right)^2 = .0034 = 0.34\%$

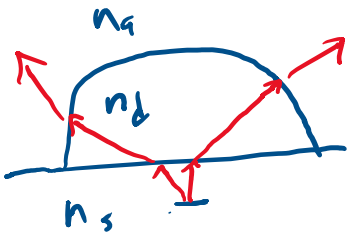
or 0.17% of all photons including downward.

c) To increase efficiency, the LED described in (a) is thinned from the bottom surface until the total thickness is 12 μm. For the photons emitted downwards, within the angular range from normal θ_m determined above, estimate the fraction absorbed between the active region and the top surface. Assume an attenuation coefficient of 0.7×10^3 cm^{-1}, and that the mirror has perfect reflectivity. State any other assumptions or approximations made. [6]



photons travelling downwards have to travel 6+12 = 18 μm to reach the top surface if at normal incidence.
Maximum angle away from normal is $\frac{1}{3.6}$ rad ≈16° so distance is not increased much ($\frac{1}{\cos 16°} = 1.04$)
$f \simeq \exp(-18\times10^{-6} \times 0.7 \times 10^5\ m^{-1}) = 0.284$
The rest, ≈ 71%, are absorbed.

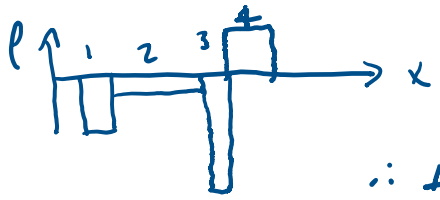
d) A domed plastic layer placed on the semiconductor surface can be used to increase η_{ext}. Why is such a structure unlikely to increase the amount of light that can be coupled from the LED into a single mode optical fibre? [4]



The dome increases the range of angles that can cross the first boundary since $n_d > n_a$ and unlike a flat piece of plastic the advantage is not lost at the 2nd boundary. Some lensing may also be achieved. However, as can be seen the area of emission from the dome will be much wider than the emitting section of the LED, and this will likely be much wider than the core in SMF.

5. A silicon avalanche photodiode (Fig. 5.1) has intrinsic and avalanche layer thicknesses of w_i and w_A respectively, acceptor doping levels N_{A1}, N_{A2} and N_{A3} in the p^+, intrinsic and avalanche layers respectively, and donor doping level N_{D4} in the n layer.

a) Assuming that a sufficient bias voltage has been applied to fully deplete both the intrinsic and avalanche regions, derive expressions for the difference in electric field strength ΔE_i across the intrinsic region, and the difference in electric field strength ΔE_A across the avalanche region, in terms of the doping levels and layer thicknesses. [4]



$$\Delta E = \frac{\rho}{\varepsilon} N_A x$$

$$\therefore \Delta E_i = -\frac{\rho}{\varepsilon_r \varepsilon_0} N_{A2} w_i$$

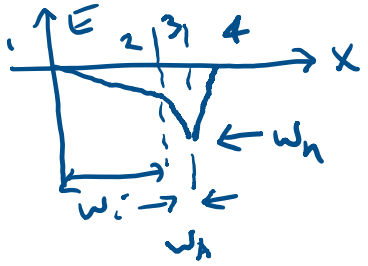
$$\Delta E_A = -\frac{\rho}{\varepsilon_r \varepsilon_0} N_{A3} w_A$$

b) Calculate the minimum bias voltage, V_{min}, to fully deplete the intrinsic and avalanche layers for the following parameter values:

$w_i = 6\ \mu m$, $w_A = 3\ \mu m$

$N_{A1} = 10^{21}\ m^{-3}$, $N_{A2} = 10^{20}\ m^{-3}$, $N_{A3} = 5\times10^{21}\ m^{-3}$ and $N_{D4} = 4\times10^{21}\ m^{-3}$

Also, calculate the width w_n of the depleted part of the n layer in this case. [6]



At V_{min}, p^+ layer is not depleted, $E(0) = 0$

$E(w_i) = \Delta E_i$

$E(w_i + w_A) = \Delta E_i + \Delta E_A$

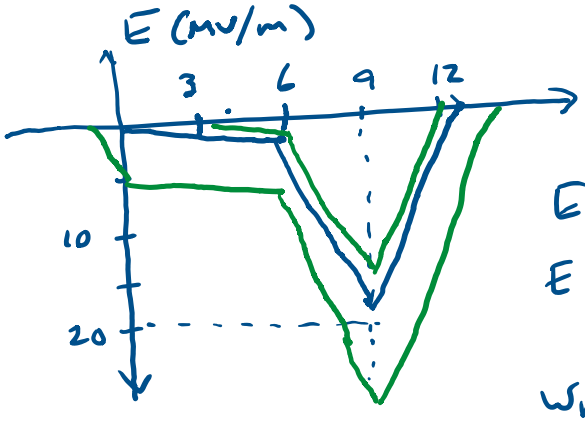
In this case get w_n using conservation of charge: $w_i N_{A2} + w_A N_{A3} = w_n N_{D4}$

$$\Delta E_i = \frac{-1.6\times10^{-19}}{12\times8.85\times10^{-12}} \times 10^{20} \times 6\times10^{-6} = -9.0\times10^5\ V/m$$

similarly $\Delta E_A = -1.75\times10^7\ V/m$

and $w_n = \dfrac{6\times10^{20} + 3\times5\times10^{21}}{4\times10^{21}} = 3.9\ \mu m$

c) Plot and dimension the electric field profile E(x) for this case where the bias voltage equals V_{min}. On the same graph, sketch additional field distributions for bias voltages higher than and lower than this value. [6]



$E(x=6) = \Delta E_i = -0.9 \times 10^6 \frac{V}{m}$

$E(x=9) = \Delta E_i + \Delta E_A$

$= -18.3 \times 10^6$ V/m

$W_n = 3.9$ μm.

For lower and higher V, the slope dE/dx in each region remains the same.

d) Discuss some factors to be considered in choosing the width of each of the intrinsic and avalanche layers in an avalanche photodiode. [4]

For the intrinsic layer:
too short will mean too few photons are absorbed, meaning too many absorbed in avalanche layer (which can make the process unstable) or lost into the substrate.
too wide means the drift time can be too long, reducing the bandwidth.
For the avalanche layer:
too short and the avalanche gain factor M will be too small. Too large and it will be hard to prevent avalanche runaway since there will be more hole-generated ionisations.